Exp 14: Pro-Code Extension using Zoho Catalyst

Aim

To study how pro-code development using Zoho Catalyst can extend the functionality of a low-code application developed in Zoho Creator and understand the integration of custom code with low-code applications.

Introduction

Low-code development platforms allow users to build applications using visual interfaces, drag-and-drop components, and predefined functions. However, complex business requirements may require additional programming and customization.

Zoho Catalyst is a pro-code development platform that provides tools for developing, deploying, and managing applications using code. It supports server-side logic, APIs, database operations, and integration with external services. Combining Zoho Creator with Zoho Catalyst can help developers extend an application's functionality beyond the features available through visual development.

Procedure

1. Log in to Zoho Creator and open the Hotel Room Booking System.
2. Review the forms, reports, dashboards, and workflows created using low-code development.
3. Open the Zoho Catalyst website and explore the available development features.
4. Study the purpose of serverless functions and understand how they execute application logic without requiring developers to manage the underlying servers.
5. Explore how APIs allow different applications and services to exchange data.
6. Understand how custom code can be used to implement advanced calculations, validations, and business rules.
7. Study how a Catalyst function could support additional features in the Hotel Room Booking System, such as automated booking validation or external payment-service integration.
8. Examine how application data could be exchanged between Zoho Creator and an external service through authenticated API requests.
9. Compare the visual development features of Zoho Creator with the code-based development approach of Zoho Catalyst.

10. Identify situations where pro-code development may be more suitable than low-code development.
11. Summarize the benefits of extending a low-code application with custom programming and external services.

Role of Zoho Catalyst in the Hotel Room Booking System

Zoho Catalyst can potentially extend the Hotel Room Booking System in the following ways:

1. **Advanced booking validation:** Custom code could validate booking dates, room availability, and other business rules.
2. **API integration:** APIs could connect the booking application with external services.
3. **Automated processing:** Serverless functions could execute specific tasks when invoked by the application.
4. **Data processing:** Custom logic could process booking details and generate additional summaries.
5. **External service integration:** The application could be extended to communicate with payment or notification services, subject to the required configuration and authentication.

These are possible extensions and do not imply that the integrations were implemented during this experiment.

Comparison Between Low-Code and Pro-Code Development

Low-Code Development	Pro-Code Development
Uses visual builders and drag-and-drop components.	Uses programming languages and code editors.
Requires less manual coding for common features.	Requires coding knowledge for custom functionality.
Suitable for rapidly developing standard business applications.	Suitable for advanced customization and complex requirements.
Provides predefined components and workflows.	Offers greater flexibility over application logic.
May have limitations for specialized requirements.	Can implement specialized functionality when supported by the platform.

Advantages of Pro-Code Extension

- Supports advanced customization beyond standard visual components.
- Enables integration with external applications through APIs.
- Provides greater control over complex business logic.
- Allows developers to build specialized application features.
- Helps extend existing low-code applications without necessarily rebuilding the entire application.

Limitations

- Requires programming knowledge and technical expertise.
- API integrations may require authentication and additional configuration.
- Custom code needs testing, debugging, and maintenance.
- Deployment and security must be handled carefully.
- Advanced integrations may require additional setup and platform resources.

Result

The pro-code extension capabilities of Zoho Catalyst were studied in relation to the Hotel Room Booking System. The study covered serverless functions, API integration, custom business logic, and the differences between low-code and pro-code development. It demonstrated how pro-code techniques can potentially extend the capabilities of a low-code application.